ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** iron	Iron is magnetic.
2	**(b)** horseshoe	U-shaped.
3	**(d)** iron nail	Iron.
4	**(a)** plastic	Plastic is non-magnetic.
5	**(c)** wood	Wood.
6	**(c)** coin of aluminium	Aluminium is non-magnetic.
7	**(c)** two poles	North and south.
8	**(b)** steel pin	Steel is magnetic.
9	**(a)** sand	Sand is non-magnetic.
10	**(a)** north and south	N and S.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Iron (1); Nickel (1).	1+1
12	They attract (1); Example: N and S (1).	1+1
13	Store in pairs with keepers (1); Avoid heat and hammering (1).	1+1
14	Plastic (1); Wood (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Iron nail (1); Steel pin (1); Paper clip (1).	1+1+1
16	Sprinkle iron filings (1); Filings cluster at the ends (1); The ends are poles (1).	1+1+1
17	North-south (1); North end points north (1); South end points south (1).	1+1+1
	OR: Plastic spoon (1); Wooden stick (1); Eraser (1).	1+1+1
18	They repel (1); Move apart (1); Like poles repel (1).	1+1+1
	OR: Parts stick together (1); Easy to join (1); Gives fun (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Magnetic: nail (1); Magnetic: pin (1); Non-magnetic: wood (1); Non-magnetic: plastic (1); Non-magnetic: rubber (1).	1+1+1+1+1
	OR: Keep in pairs (1); Use keepers (1); Avoid heat (1); Avoid hammering (1); Keep from gadgets (1).	1+1+1+1+1
20	Take objects (1); Bring magnet near each (1); See which stick (1); Note in a table (1); Conclude (1).	1+1+1+1+1
	OR: Hang one magnet (1); Bring N near N (1); They repel (1); Bring S near N (1); They attract (1).	1+1+1+1+1

Mistakes to avoid:

- Thinking everything metallic is magnetic; copper and aluminium are metals but **not** magnetic.
- Using attraction alone as the test for a magnet (iron is attracted too); the sure test is **repulsion**.
- Writing 'like poles attract'; **like poles repel** and unlike poles attract.
- Thinking the centre of a bar magnet is the strongest part; it is strongest at the **poles**.
- Believing a magnet cut in half gives a separate north and south pole; each piece has both.
- Forgetting that a freely suspended magnet comes to rest in the north-south direction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** poles	Poles.
2	**(b)** north-south direction	North-south.
3	**(d)** repel	Same poles repel.
4	**(d)** poles	Poles.
5	**(c)** geographic north	North-seeking.
6	**(b)** attract	Opposite attract.
7	**(b)** two magnets	Each piece is a magnet.
8	**(c)** keepers	Keepers help.
9	**(c)** magnet	It is magnetised.
10	**(c)** compass	Compass.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Instrument to find direction (1); Has a magnetic needle (1).	1+1
12	A natural magnet (1); Used by early sailors (1).	1+1
13	North pole (1); South pole (1).	1+1
14	Bar (1); Horseshoe (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Compass (1); Door catch (1); Toy (1).	1+1+1
16	They attract (1); Stick together (1); Unlike poles attract (1).	1+1+1
17	It may spoil them (1); Magnet disturbs parts (1); Keep at distance (1).	1+1+1
	OR: Has two poles (1); Attracts iron (1); Points north-south (1).	1+1+1
18	Needle (1); Dial (1); Case (1).	1+1+1
	OR: Magnetic: attracted (1); Non-magnetic: not attracted (1); Example: iron and glass (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Compass (1); Fridge door (1); Magnetic board (1); Toys (1); Cranes (1).	1+1+1+1+1
	OR: Two magnets form (1); Each has two poles (1); They still attract iron (1); Cutting does not give single poles (1); They are weaker (1).	1+1+1+1+1
20	Keep it flat (1); Wait for needle to rest (1); Needle points north (1); Turn dial N to the needle (1); Read other directions (1).	1+1+1+1+1
	OR: A stone attracted iron (1); Called lodestone (1); People hung it (1); It pointed north-south (1); Used for sailing (1).	1+1+1+1+1

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- Writing 'like poles attract'; **like poles repel** and unlike poles attract.
- Thinking the centre of a bar magnet is the strongest part; it is strongest at the **poles**.
- Believing a magnet cut in half gives a separate north and south pole; each piece has both.
- Forgetting that a freely suspended magnet comes to rest in the north-south direction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** mobile phone	May damage electronics.
2	**(c)** direction	It points north.
3	**(b)** lodestone	Lodestone.
4	**(d)** direction	Travellers.
5	**(a)** refrigerator	Fridge doors use magnets.
6	**(a)** weak	Heat weakens it.
7	**(a)** yes, the force acts through paper	Paper does not block.
8	**(b)** nail	Nails are iron.
9	**(a)** weak	Hammering weakens.
10	**(c)** iron or steel	Steel body.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Magnetic materials (1); Such as iron (1).	1+1
12	They repel (1); Example: N and N (1).	1+1
13	Compass (1); Fridge door (1).	1+1
14	Force exerted by a magnet (1); Attracts or repels (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Bar (1); Horseshoe (1); Ring (1).	1+1+1
16	Take iron piece (1); Rub with magnet (1); Always in one direction (1).	1+1+1
17	Pulling together (1); Magnet pulls iron (1); Opposite poles attract (1).	1+1+1
	OR: Do not heat (1); Do not hammer (1); Use keepers (1).	1+1+1
18	Bring magnet near (1); Observe attraction (1); Write the result (1).	1+1+1
	OR: Pushing away (1); Like poles repel (1); It tells a magnet from iron (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Take a steel bar (1); Take a magnet (1); Rub in one direction (1); Repeat (1); Test with pins (1).	1+1+1+1+1
	OR: Mix them (1); Move magnet over mixture (1); Pins stick (1); Remove pins (1); Sand stays (1).	1+1+1+1+1
20	Attracts iron (1); Two poles (1); Like poles repel (1); Unlike attract (1); Points north-south (1).	1+1+1+1+1
	OR: Spread filings on a sheet (1); Roll the magnet in them (1); Lift the magnet (1); Filings cling at the ends (1); Ends are poles (1).	1+1+1+1+1

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